



Technical Service Bulletin

Technical Service Bulletin: TSB190206

Released Date: 09-Dec-2019

New OEM Software to Prevent Engine Brake Rocker Lever Assembly Wear on Epiroc's MT-42 Unit

New OEM Software to Prevent Engine Brake Rocker Lever Assembly Wear on Epiroc's MT-42 Unit

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Information Only - OEM Related Matter Not Covered By Cummins® - Contact Appropriate OEM Dealer or OEM Representative For Additional Information

Contents

Product Affected

Engine:

- QSX15 CM570

Original Equipment Manufacturer (OEM):

- Epiroc

Chassis:

- MT-42 Mining Truck

Issue

Symptom:

- Units may experience noise or vibration coming from the engine.
- Engine brake parts, including shaft detent, control valve springs and control valve covers will appear worn out and become loose from their assembly.

Root Cause:

- TSC1 Torque Control message from OEM software arriving on the J1939 datalink configured incorrectly. The incorrect configuration causes the Torque Control messages to drive excessive engine brake activations up to several times the nominal activations seen on these applications.

Confirmation

- Inspect the back of the engine brake switch in the dashboard for any abnormal wiring. All wires should be directed to their respective ports. See equipment manufacturer service information.
- Perform a complete overhead inspection of the engine brake parts. Take apart the rocker lever/shaft assembly and inspect the following:
 - Shaft detent: Detent will show rough wear around the edges as a result of the control valve unable to sit permanently. See Figure 1 below.
 - Control valve springs: Springs will be shortened in comparison to otherwise healthy or new springs. See Figure 2 below.
 - Rocker levers brake actuator piston: Piston will show rough wear around the edges. See Figure 3 below.
 - Engine brake contact bridge pin: Bridge pin will show rough wear around the edges. See Figure 4 below.
 - The figures below show examples of worn out components.



Figure 1, Engine Brake Shaft Detent Hole.



Figure 2, Engine Brake Control Valve Spring.



Figure 3, Engine Brake Actuator Piston.



Figure 4, Engine Brake Contact Bridge Pin.

Resolution

- Contact the OEM for further instructions on how to address this type of failure.

Document History

Date	Details
2019-11-22	Module Created

Last Modified: 09-Dec-2019